APGM — Adaptive Perception Governance Module

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Canonical Definition

Adaptive Perception Governance Module defines the methodological conditions under which autonomous systems analyze perception failure, adjust interpretation priority, govern sensor weighting, and improve physical-world interpretation without losing structural control, traceability, or validity.

A system satisfies APGM only if:

- perception failure can be analyzed as a governed event
- adaptation remains bounded by defined rules
- sensor priority logic is explicit
- learning or correction remains traceable
- physical execution is restricted when adaptation logic is absent, unreliable, or ungoverned

A system that changes perception behavior without governed priority and traceable correction does not satisfy APGM.

Module Function

APGM defines the adaptive governance layer of physical reality interpretation.

It ensures that systems do not merely sense and interpret physical conditions, but can also:

- identify why interpretation failed
- determine which sensing source should be prioritized
- adjust interpretation under governed conditions
- improve future reliability without uncontrolled self-modification

The module applies wherever autonomous systems must learn from collision, near-miss, false positive, false negative, recurring ghost event, or environmental misinterpretation.

Minimum Implementation Framework (MIF)

Step 1 — Define Perception Failure Conditions

The organization must define what counts as perception failure.

Minimum requirement:

- collision, near-miss, false trigger, missed detection, or persistent distortion conditions are identifiable
- failure categories are explicit
- undefined failure is excluded from valid adaptive logic

Step 2 — Define Failure Source Analysis

The system must define how the source of perception failure is analyzed.

Minimum requirement:

- the analysis path is traceable
- sensor, environment, interpretation logic, or execution context can be examined
- failure is not reduced to unexplained anomaly only

Step 3 — Define Sensor Priority Logic

The system must define which sensing source has priority and under what conditions that priority may change.

Minimum requirement:

- priority rules are explicit
- changing priority conditions are identifiable
- arbitrary or hidden source preference is excluded

Step 4 — Define Adaptive Correction Boundary

The system must define how perception adaptation may occur.

Minimum requirement:

- adaptation is bounded
- learning does not silently rewrite validity conditions
- correction logic remains inside governed limits
- uncontrolled self-adjustment is excluded

Step 5 — Preserve Traceable Learning Path

The system must preserve traceability of correction and adaptation.

Minimum requirement:

- changes in interpretation logic are reviewable
- correction path remains visible
- later behavior can be linked to prior failure analysis
- learning is not treated as valid if its basis cannot be reconstructed

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Step 6 — Restrict Invalid Execution

The system must not be treated as valid if physical execution proceeds while adaptive perception logic is ungoverned, unexplained, or structurally unreliable.

Minimum requirement:

- invalid adaptive conditions are identifiable
- unresolved failure source blocks valid reliance where required
- learning alone does not restore interpretation validity
- adaptation does not override safety and integrity conditions

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Use Case 1 — Repeated False Trigger in Fixed Environment

Scenario

An autonomous system repeatedly interprets a phantom object, obstacle, or movement event at the same location, although no real operational threat exists.

Application

APGM is used to ensure:

- the repeated failure is recognized as a governed perception event
- the source of false interpretation is analyzed
- sensor priority and interpretation weighting are adjusted under explicit rules
- future execution does not rely on uncontrolled learning or silent normalization

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Result

The organization gains a structured way to improve perception reliability without allowing repeated ghost behavior to silently distort execution logic.

Use Case 2 — Collision or Near-Miss Analysis in Autonomous Operation

Scenario

A robot, vehicle, or embodied system experiences collision, near-miss, or unsafe physical behavior and must determine why perception failed and which sensing or interpretation condition should have been prioritized.

Application

APGM is used to ensure:

- the failure source is analyzed
- priority between sensing sources is explicit
- corrective adaptation remains bounded and traceable
- future execution remains governed by defined interpretation logic rather than uncontrolled self-modification

Result

The organization gains a stronger basis for perception improvement, safer future operation, and more defensible governance of learning-based correction in physical environments.

Canonical Closing Statement

If a system adapts perception without governed priority, bounded correction, and traceable learning, physical interpretation is not structurally reliable.

Related Documents

→ Physical Reality Interpretation Layer (PRIL™)

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Canonical Source

https://originopenfoundation.org/